

Released-Materials Reproduction, Study 1 (or1-)

Running the Anthropic Jacobian-lens battery on Qwen 3.6 27B and OLMo 3.1 32B Instruct development/methods tier — prompt-exact where released, method-reconstructed where specified, gated where not identified

J-space campaign — official-repro sideline

branch `interp_jspace_official_repro_1` — upstream `anthropics/jacobian-lens @ 581d3986`

2026-08-08 (OR1-A Qwen block)

Abstract

We run the released Jacobian-lens evaluations (6 sets) and experiment prompt sets (11 sets) from the public companion release of “Verbalizable Representations Form a Global Workspace in Language Models” under one audited harness on two open-weight models, and separate prompt, lens, intervention, capability, and model effects. This report covers the Qwen lane: the campaign-pinned checkpoint with the published $n=1000$ lens — the study’s mechanical instrument anchor (OR-Q1) and its first generalization surface (OR-Q2). Fidelity classes and gate states are carried on every cell; gated cells are missing, never zero; no confirmatory language is licensed anywhere in Study 1.

1 Provenance and instrument identity

Upstream release	<code>anthropics/jacobian-lens @ 581d3986</code> (11 experiments + 6 evals, hash-pinned)
Qwen checkpoint	<code>Qwen/Qwen3.6-27B @ 6a9e13bd</code> (64 layers, $d=5120$, 48 GDN blocks verified)
Published lens	<code>neuronpedia/jacobian-lens @ a4114d7</code> , $n=1000$, sha256 1718c8c5... byte-verified
Readout parity	max abs. diff 0.0e+00 (hard stop threshold 10^{-4} ; exact top-20 order agreement)
$\alpha=0$ no-op	exact (0.0 logit difference across a full-position band session)
g-folding audit	min cosine 0.382 over 26 battery tokens $\times$ 14 band layers — material ; D9: paper-literal unfolded v_t stays primary
Bands	paper grid 25 layers; workspace band [24, 58]; campaign band sensitivity [20...44]

Table 1: Lane admission. The boot-probe sentinel (Fig. 1) reads the latent country mid-band and the currency near the output — the instrument behaves as the paper describes before any experiment runs.

2 The paper’s own example, on Qwen

3 Six released lens evaluations (OR-Q1)

4 Qwen causal core (OR-Q2)

4.1 Verbal report

Release-literal rule (first ten listed candidates, skip the answer), swap-out = greedy token at the generation boundary (D1), $\alpha=1$ across the band at every prompt position. Category-equal top-1 0.0% and top-5 0.8% over 14 categories (118 executed trials; 64 trials improved the candidate’s rank, 53 worsened). Paper context (Claude): 88% top-5. The effect on Qwen under the paper-literal operation is *directionally present in rank movement but far below the paper’s report-rate*: Fig. 5.

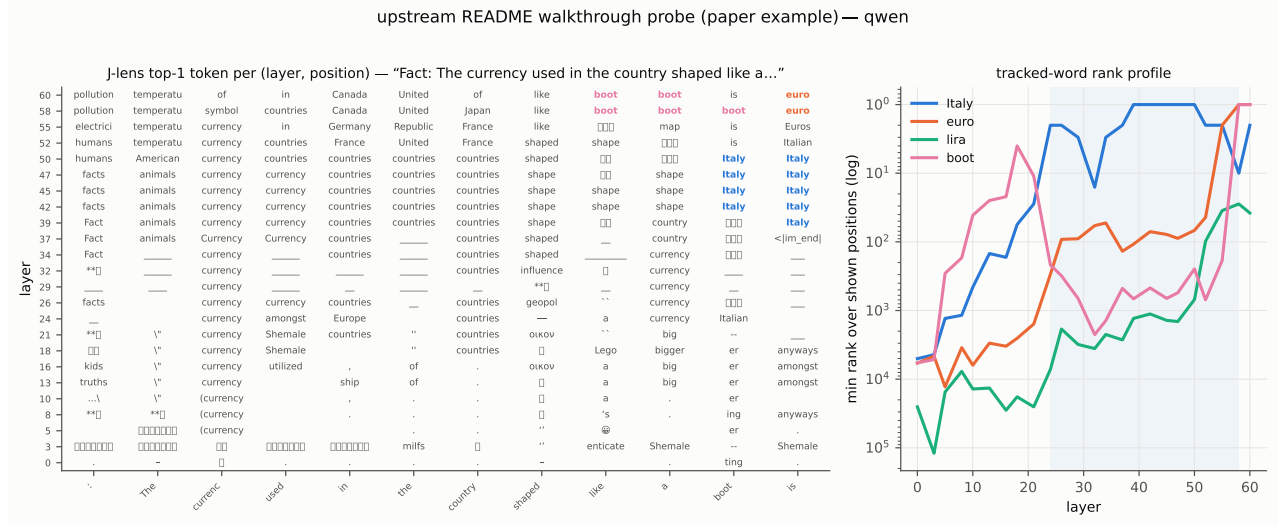


Figure 1: The upstream README walkthrough probe (“The currency used in the country shaped like a boot is”) under the published lens on Qwen 3.6 27B. Left: J-lens top-1 token per (layer, position); colored cells are tracked words. Right: tracked-word best rank by layer (shaded: workspace band). *Italy* appears as latent content mid-band and hands off to *euro*/*lira* toward the output — the paper’s slice-visualisation phenomenology, reproduced from released materials.

Set	Jacobian lens			logit lens		
	@1	@5	@20	@1	@5	@20
multihop ($n=93$)	0.244	0.399	0.641	0.173	0.327	0.542
multilingual ($n=107$)	0.237	0.384	0.580	0.184	0.327	0.497
poetry ($n=98$)	0.010	0.020	0.071	0.020	0.031	0.061
typo ($n=96$)	0.135	0.271	0.500	0.333	0.594	0.719
order-ops ($n=55$)	0.145	0.455	0.727	0.209	0.600	0.727
association ($n=102$)	0.039	0.059	0.108	0.010	0.059	0.118

Table 2: pass@ k , token-valid primary. All-source-layer sensitivity and released-denominator sensitivity live in the registered JSON.

4.2 Flexible generalization

All 192 ordered off-diagonal swaps; capability-conditioned primary (source and target baselines correct; baseline accuracy 51.6%); $\alpha=2$ full sensitivity. Category-equal capable top-1: 5.8% ($\alpha=1$) $\rightarrow$ 0.0% ($\alpha=2$); diagnostic all-executable: 5.6%. Paper context: 76/192 $\rightarrow$ 101/192 (Sonnet 4.5). Per-category grids in Fig. 6.

4.3 Probe-swap, raw J-lens token arm

Fidelity `prompt_exact_representation_adapted_raw_jlens` (the paper’s own §3.3 headline arm; the official linear-probe arm is NOT-IDENTIFIED — its training corpus is absent from the release). Population: 90 released items, 7 tokenization-gated, 83 executed, 63 baseline-capable. Capable top-1 14.3% (diagnostic 14.5%); multihop 10.5% vs non-multihop 15.9%. Paper context: 60% (Claude, $n=90$).

5 Interpretation under the wording router

Conformance and admission passed mechanically (exact parity, exact no-ops, verified pins), so every result above is a scientific observation about this model/instrument cell, not a harness state. The lens-eval layer profiles reproduce the paper’s methods-comparison direction on the latent-content distributions; the paper-literal $\alpha=1$ coordinate-swap core is *directionally present but substantially weaker on Qwen 3.6 27B than the paper’s Claude values*, with rank movement concentrated below top-1. Maximum licensed sentence (plan

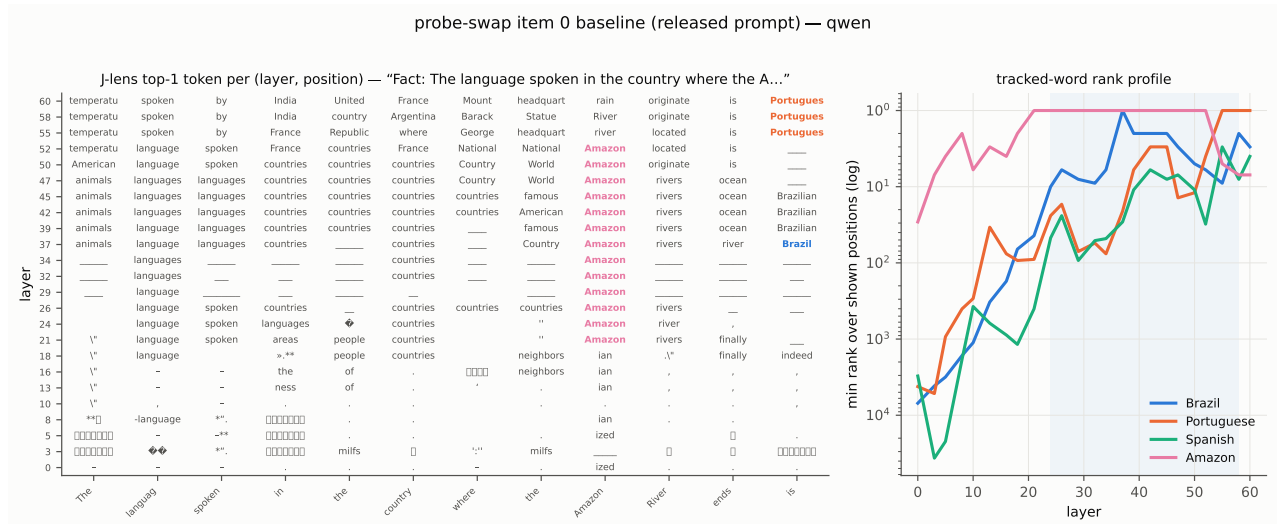


Figure 2: Released probe-swap item 0 (“the language spoken in the country where the Amazon River ends”) — the two-hop structure is visible: **Brazil** (bridge entity) carried in-band while the output side resolves toward **Portuguese**.

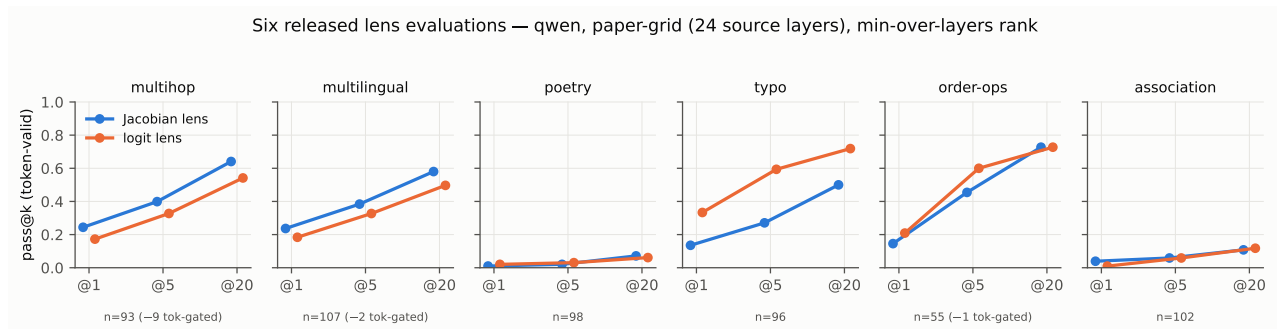


Figure 3: pass@ k per released set, token-valid denominators, min-over-24-paper-grid-layers at the released readout position.

§7.6 routing, pre-OLMo): *“conformance passes; the Qwen core is a valid generalization result”* — the released functional properties do not transfer at paper magnitude to this cell under prompt-exact, method-reconstructed conditions. No workspace-existence or -absence claim is licensed; OLMo, the extended battery, and the instrument cross-over (which will separate intervention semantics from prompt population) follow in OR1-B/C/D.

6 Population accounting

Every released row carries source-present / tokenization-valid / geometry-valid / baseline-capable / executed states in the registered raw records (`*_raw.jsonl`, plan §12 schema); the attrition waterfall figure ships with the OLMo lane so both lanes share one accounting frame. Nothing gated was encoded as zero anywhere in this report.

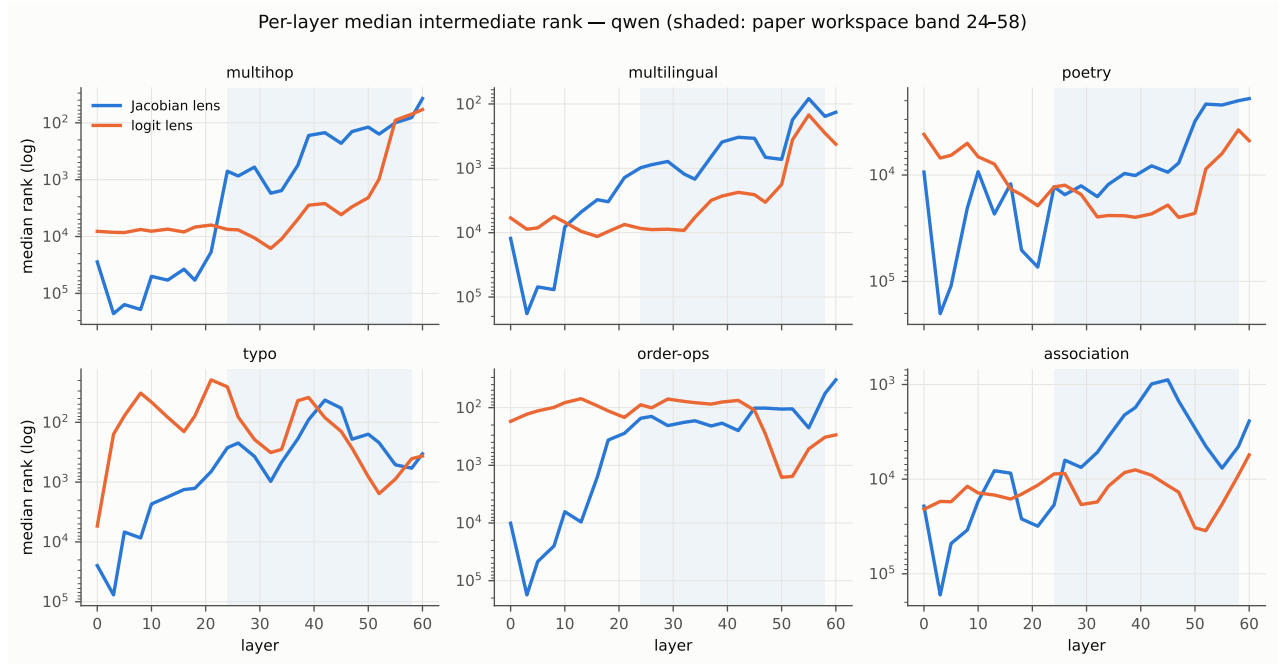


Figure 4: Median intermediate rank by layer. On the latent-content sets (multihop, multilingual) the J-lens reaches useful ranks by the band onset while the logit lens remains near rank 10^4 until $\sim L45$ — the paper’s central methods-comparison claim, visible on an open model. On typo the logit lens is stronger (the correction is surface-adjacent); order-ops numbers sit in the prompt, so the logit lens copies them early.

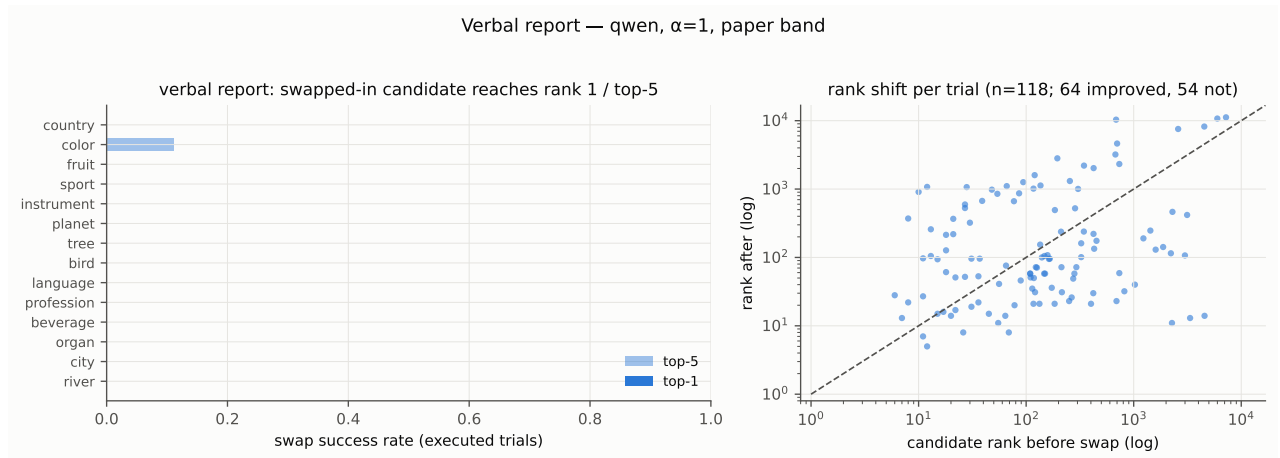


Figure 5: Verbal report on Qwen: per-category success rates (left) and per-trial rank shifts (right; below the diagonal = swap moved the candidate toward the output).

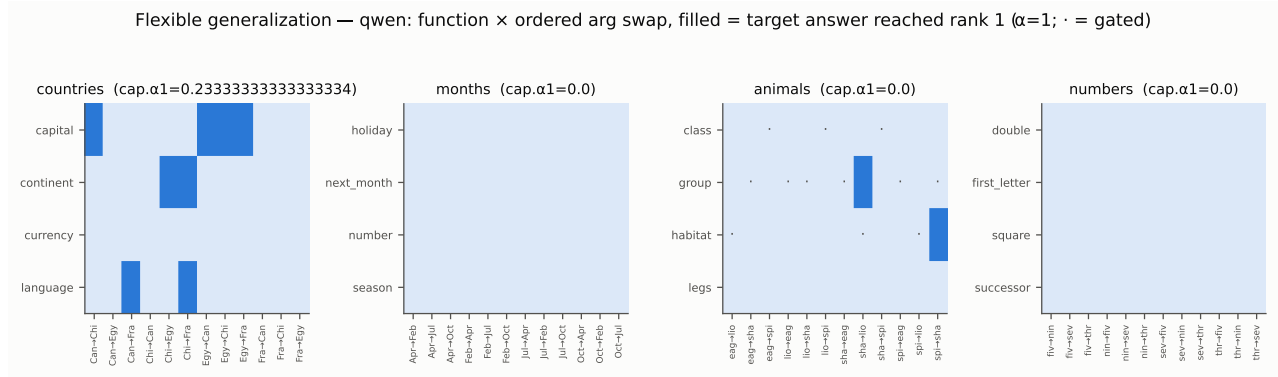


Figure 6: Function × ordered-argument-swap outcomes, $\alpha=1$ (filled = target answer reached rank 1).

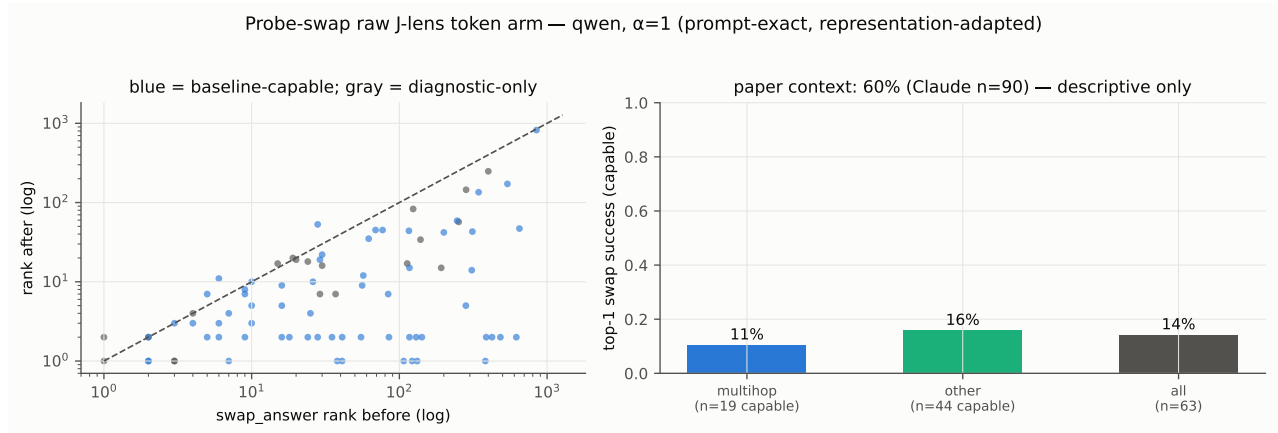


Figure 7: Probe-swap raw arm on Qwen: per-item rank movement of the predicted swapped answer (left) and capable top-1 by category group (right).